1 Similarity

Last time we saw that if $T \in \mathscr{L}(V)$ for a finite-dimensional vector space V and B, B' are ordered bases of V, then the matrices of T with respect to these bases are related by

$$[T]_{B'} = C^{-1}[T]_B C,$$

where C is the transition matrix from B to B'.

Definition 1.1 (Similar matrices)**.** Two $n \times n$ matrices A and B are said to be *similar* if there exists an invertible matrix C such that $B = C^{-1}AC$.

Exercise 1.2. Prove that similarity is an equivalence relation on $M^{n\times n}(F)$.

Definition 1.3 (Similarity invariants)**.** A property of an $n \times n$ matrix A which holds for all matrices similar to A is called a *similarity invariant*.

Exercise 1.4. Show that invertibility is a similarity invariant.

Name:_____

Exercise 1.5. Show that the nullity and rank of a square matrix are similarity invariants. (*Hint*: If $B = C^{-1}AC$ and $\{v_1, \ldots, v_k\}$ is a basis of Nul A, show that $\{C^{-1}v_1, \ldots, C^{-1}v_k\}$ is a basis of Nul B.)

Remark 1.6. If $B = C^{-1}AC$, then $B^T = C^TA^T(C^{-1})^T = C^TA^T(C^T)^{-1} = ((C^T)^{-1})^{-1}A^T(C^T)^{-1}$, which shows that B^T and A^T are similar. It follows from Exercise 1.5 that B^T and A^T have the same rank and nullity. Thus, if A and B are similar, then each fundamental matrix space of A has the same dimension as the corresponding fundamental matrix space of B.

Exercise 1.7. Prove that the matrices $A = \begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ are not similar.